

AC9M3A02 • Year 3 Mathematics

Efficient Mental Addition and Subtraction

Known facts such as doubles, complements to 10 and fact families can be extended to tens and hundreds for efficient mental calculation.

We are learning to...

extend and apply knowledge of addition and subtraction facts to 20 to develop efficient mental strategies for computation with larger numbers without a calculator

Recall known fact

Adjust place
value

Choose strategy

Calculate
mentally

Check

Success criteria

- I can extend facts to tens.
- I can use bridging.
- I can use compensation.

Key vocabulary

mental strategy

compensation

friendly number

Extend $7 + 8 = 15$

Known fact	Extended fact
$7 + 8 = 15$	$70 + 80 = 150$
$15 - 8 = 7$	$150 - 80 = 70$

The digits keep the same relationship while place value changes.

Use compensation and bridging

$$198 + 47$$

→

$$200 + 47$$

→

$$247 - 2$$

→

$$245$$

Adjust to a friendly number, then compensate.

Show what you know

Use $7 + 8$ to solve $70 + 80$.

Think silently, show your reasoning, then explain it to a partner.

Expected response

$70 + 80 = 150$.

If students are unsure

Use a known fact or place-value relationship.

mental strategy

compensation

friendly number